

# W4L1 | Counter crossings

Lesson 1 of Week 4 | 40 minutes | Introduce | Editable pupil sheet

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Core first. Write, speak, point or dictate your own words. Keep the first attempt so you can improve it.

**Optional support:** Net movement = crossings one way minus crossings the other way. Start: “More particles move from \_\_\_ to \_\_\_ because...”

## Work together

Draw two equal-volume regions: 12 particles left and 3 right. Predict the initial net direction.

Illustrative model/sample data; not measurements made by this class.

| Model interval | Left → right | Right → left |
|----------------|--------------|--------------|
| Early          | 8            | 2            |
| At balance     | 5            | 5            |

**Your labelled sketch / arrows:**

1. Use the early crossing counts to calculate net movement and direction.

2. At balance, how much net movement is there? What is still happening?

# W4L1 | Continue and improve

Use after the core questions when directed

3. Explain one reason a higher temperature can increase diffusion rate.

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## Optional transfer

Why must the two starting regions have equal volumes if particle counts alone are used to compare concentration?

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Depth: name an assumption or limit, then say what evidence would strengthen your answer.

## Improve the same work

One correction I made, and why:

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My request for next time: \_\_\_\_\_

# W4L1 | Arrival — choose one route

Four quick questions | Point, say, sign, draw, type or write | You may pass

Choose one route. Point, say, sign, draw, type or write. You may pass or say not sure. Use the screen or paper; you do not need to do both. Nothing here assumes you have met diffusion before. The questions use everyday ideas about smells, drinks and breathing; an adult can read them to you or write your words.

## Supported route

1. Someone opens a bottle of perfume at the front. Can people at the back smell it after a while? Yes / No.

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2. Two glasses of squash: one strong, one weak. Which one has more squash particles in the same amount of water? Strong / Weak.

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3. In a gas, are the particles close together or far apart? Close / Far apart.

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4. When you breathe in, where does oxygen get into your blood? Lungs / Stomach.

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## Standard route

1. A smell spreads across a still room. What is actually moving from the bottle to your nose?

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2. What does the word concentration mean when we talk about a solution or a gas?

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3. Particles move in solids, liquids and gases. In which state do they move most freely?

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4. Name the gas that our blood collects in the lungs and the gas it gives up there.

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## Stretch route

1. A smell reaches the back of a still room. Explain what makes the particles spread when nothing is pushing them.

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2. Two areas have different concentrations of the same substance. What name do we give to that difference?

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3. Why do particles in a warm gas spread out faster than particles in a cold gas?

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4. Oxygen passes from the air sacs of the lungs into the blood. Suggest why it goes that way and not the other.

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# W4L1 | Knowledge organiser

Keep this page; use it in any stage

Objective: Explain diffusion and the factors that affect its rate.

## Success criteria

1. Define diffusion using net movement and concentration gradient.
2. Name three factors that change the rate of diffusion and say which way each one changes it.
3. Apply diffusion to a real example in the body and to the model shown, keeping the two apart.

## Key words

| Word                   | Tier   | Meaning                                                                               |
|------------------------|--------|---------------------------------------------------------------------------------------|
| Diffusion              | Tier 3 | The net movement of particles from a higher concentration to a lower concentration.   |
| Concentration          | Tier 2 | How many particles of a substance there are in a given volume.                        |
| Concentration gradient | Tier 3 | The difference in concentration between two places. Particles diffuse down it.        |
| Net movement           | Tier 3 | The overall movement once movement in both directions has been counted.               |
| Passive                | Tier 3 | Needing no energy from the cell; diffusion is passive.                                |
| Surface area           | Tier 2 | The total area available for particles to cross.                                      |
| Membrane               | Tier 3 | A thin layer that particles must cross, such as a cell membrane.                      |
| Alveolus               | Tier 3 | One of the tiny air sacs in the lungs where oxygen enters the blood. Plural: alveoli. |
| Rate                   | Tier 2 | How fast something happens; here, how quickly particles move across.                  |

## Key ideas

Particles move randomly all the time. Diffusion is the net result of that random movement, not a push.

Three factors raise the rate of diffusion: a steeper concentration gradient, a higher temperature and a larger surface area. A shorter distance to cross also raises it.

Diffusion needs no energy from the cell. Movement against a gradient using energy is active transport, a different process.

In the lungs, oxygen diffuses from the air in an alveolus into the blood; carbon dioxide diffuses the other way. The alveolus wall is one cell thick.

A particle model of diffusion across a membrane. Particles are drawn far larger and fewer than the real thing, and the membrane is drawn as a line. This is an explanatory model, not a measurement of any real gas.

### Diffusion or not?

Diffusion: oxygen into the blood at the alveoli; carbon dioxide out of a respiring cell; a smell spreading across a room.

Not diffusion: glucose pumped into a root hair against its gradient (active transport); water across a partially permeable membrane (osmosis, a special case treated in the next lesson but one).

# W4L1 | Exit ticket — complete one route, once

Two small checks | You may pass

Two small checks to finish. Point, say, sign, draw, type or use paper. An adult can scribe your words. You may pass; complete one route, once.

## Supported route

1. Diffusion is the net movement of particles from a higher concentration to a \_\_\_\_ concentration.  
(lower / higher)

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2. A steeper concentration gradient makes diffusion happen \_\_\_\_\_. (faster / slower)

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## Standard route

1. Complete the definition: diffusion is the \_\_\_\_ movement of particles from a higher to a lower \_\_\_\_.

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2. Name two factors that increase the rate of diffusion.

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## Stretch route

1. Explain why a thinner membrane increases the rate of diffusion. Use the word distance.

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2. Explain why warming a gas increases the rate of diffusion. Use the words energy and particles.

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Next lesson: Diffusion in the Lungs: Testing Breath.

Verbal, written, drawn, signed, symbol or choice selected, demonstrated or re-attempted, adult scribed: all count.